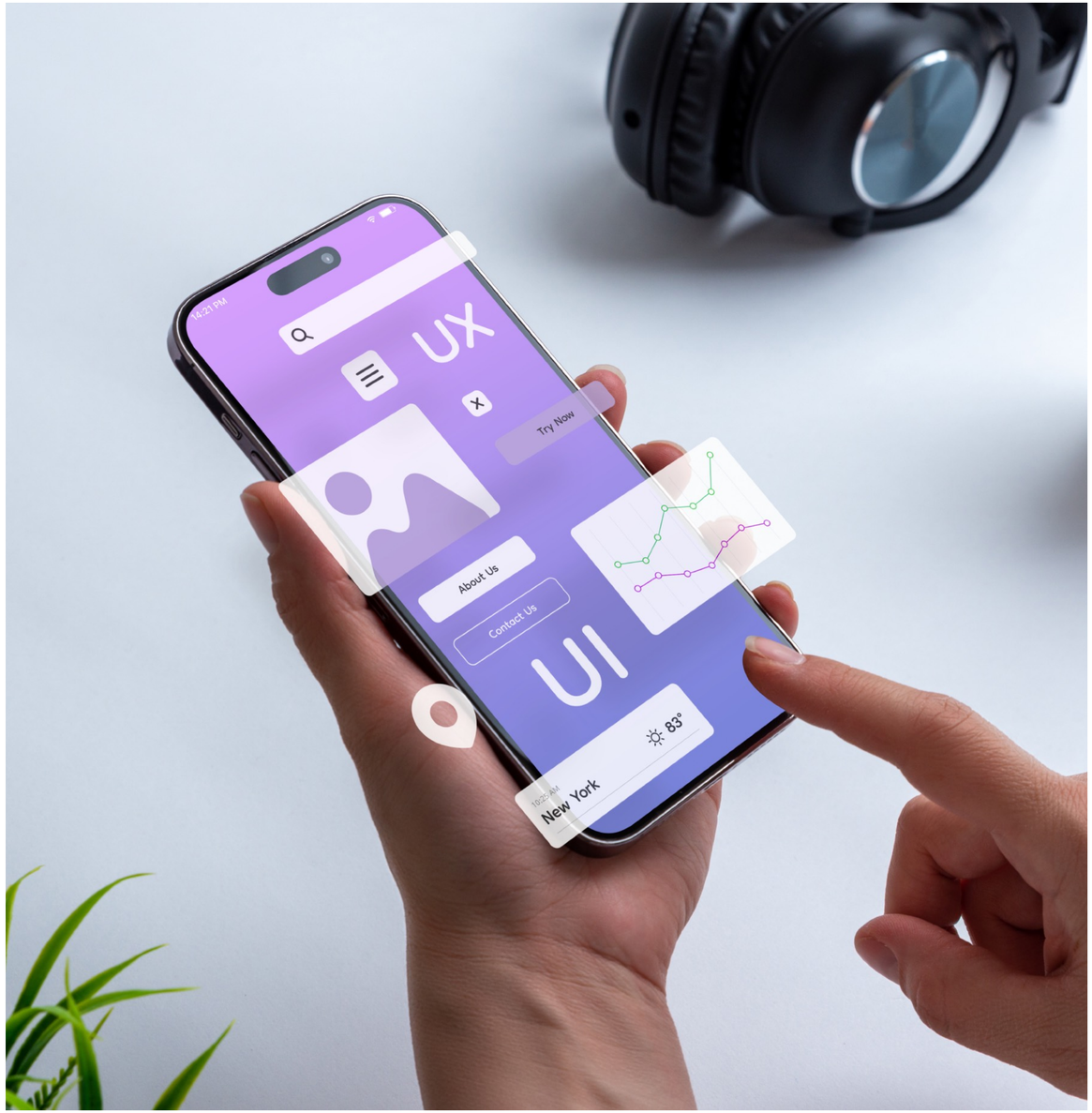
User Experience Design

Ing. Lisa-Maria Freller, MA

High-Fi prototyping

Today

- Recap visual design
- Figma: creating interactive high-fi prototypes
- High-fi prototyping with AI & Best practices
- UX management & design workflow



Recap: Best of ID

Quiz „Best of Interface Design“

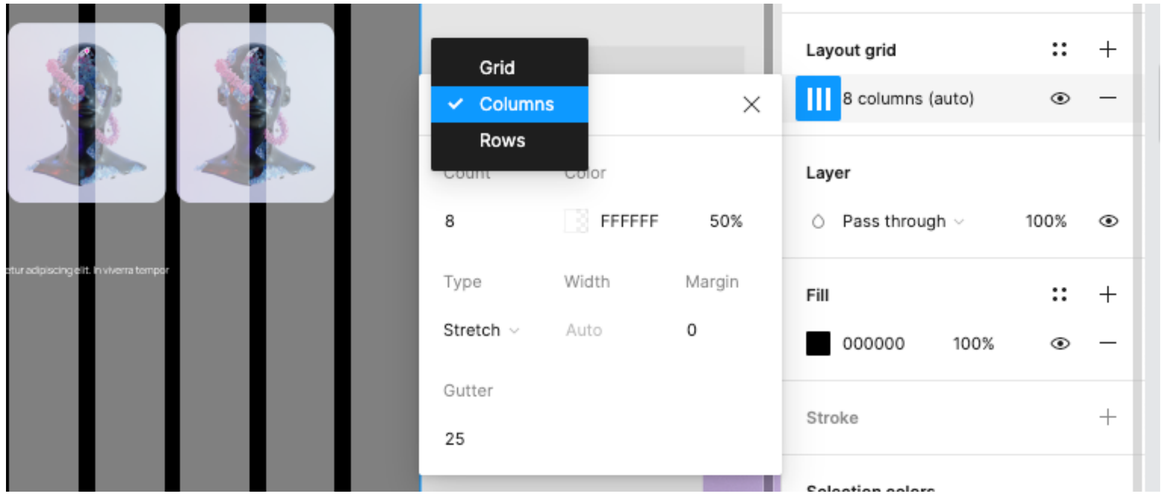


Grids & Spacing

- ✓ Frames use grids e.g. 12 cols
- ✓ Common figma structure: columns: 12, stretch, margin: 16, gutter: 8
- ✓ Show/Hide layout grid: `Ctrl+↑+4`

👉 **Benefit:** Keeps layout consistent and ensures spacing throughout design





Grids & Spacing

✓ Token example:

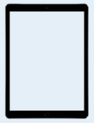
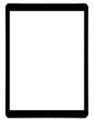
- #0066FF → Primary / Blue
- 12px → $spacing-04

✓ Instead of values use **tokens**

✓ Use **multiples of** e.g. 4 px for spacing

TOKEN	PX	REM
$spacing-01	2	0.125
$spacing-02	4	0.25
$spacing-03	8	0.5
$spacing-04	12	0.75
$spacing-05	16	1
$spacing-06	24	1.5
$spacing-07	32	2
$spacing-08	40	2.5

Grids & Spacing

	BREAKPOINT	COLUMNS	MARGIN	GUTTER
	> 1600 px	16	20	24
	1599-1200px	12	20	24
	1199-769px	8	24	24
	<767px	4	24	24

Every design has a Personality

Typeface

Elegant or classic: serif

Playful: rounded sans serif

Neutral: sans serif

Border radius

Rounded corners (2 px - 6 px) are considered friendlier than sharp

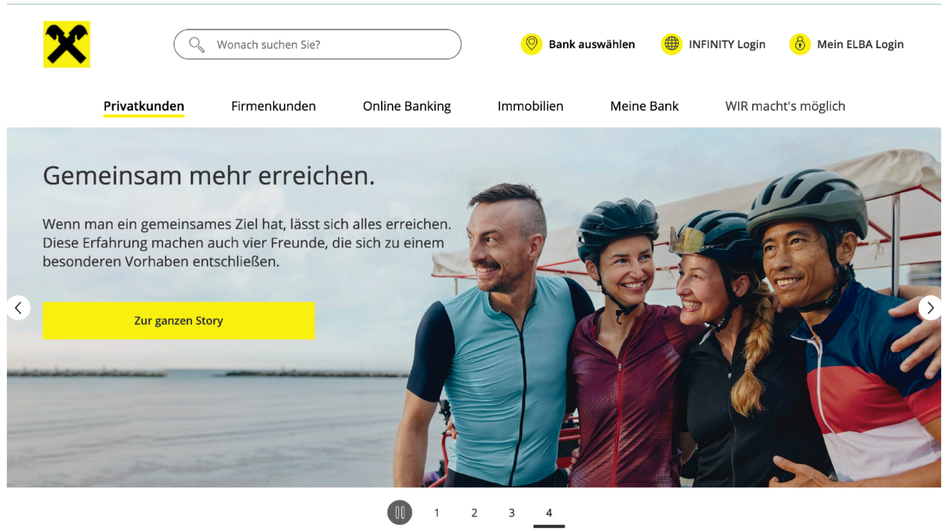
Color

Gold might say „expensive“

Pink „fun“

Language

Defines voice and tone



Banking: secure, professional

Every design has a Personality

- ✓ Limit your choices
- ✓ Set your colors and typeface
- ✓ Define a sizing scale

Moodboard

Establish a visual identity

Moodboard • Define the overall mood and emotions the interface should transfer

Stylescape • Moodboards taken to „next level“ including detailed definition of visual style:

- Graphics, Photos, Illustrations
- Color Schemes
- Fonts
- Iconography



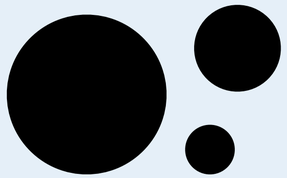


Visual Design Principles

... help create visually consistent, engaging, and usable interfaces

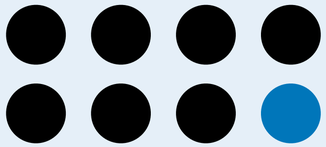
Scale

Refers to using relative size to signal importance and rank in a composition



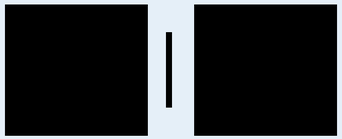
Contrast

Placing visual elements next to each other to visualise differences



Balance

Visual elements are evenly distributed so the design doesn't feel tilted or heavy on one side



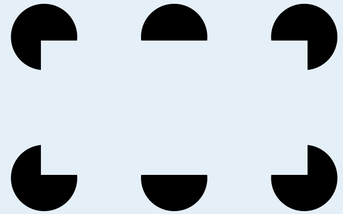
Hierarchy

Guiding the eye on the page in the order of importance



Gestalt

Tendency to perceive the whole as opposed to the individual elements

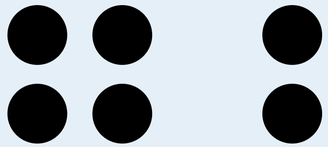


Gestalt Principles

… explain how people see visual elements as organised wholes rather than isolated parts - and provide psychological foundation for visual design decisions

Proximity

Close elements are perceived as group



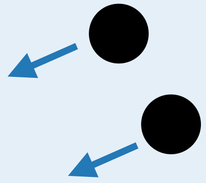
Similarity

Similar elements are related



Common Fate

Moving together = grouped



Closure

Brain completes missing parts



Figure Ground

Separate focus from background



Design System

Resources

Sample page

Brand guidelines

Developer documentation

Design principles

Ux recommendations

Themes & architecture

Style Guide

Design guidelines

Icon

Typography

Brand

Color

Margins & Paddings

Pattern Library

Reusable **UI & interaction solution**

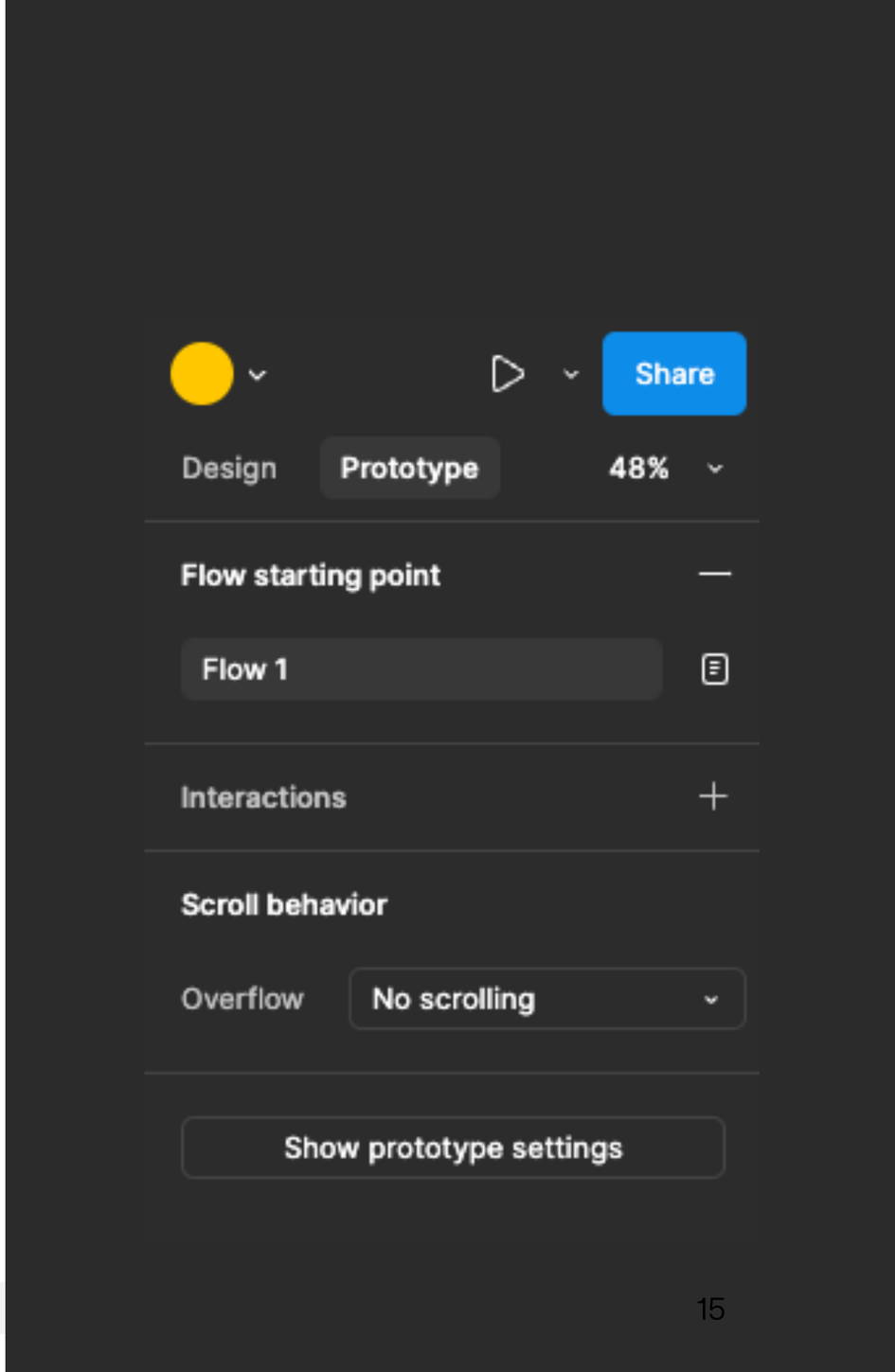
Component Library

Code snippets

Creating High-Fi Wireflow

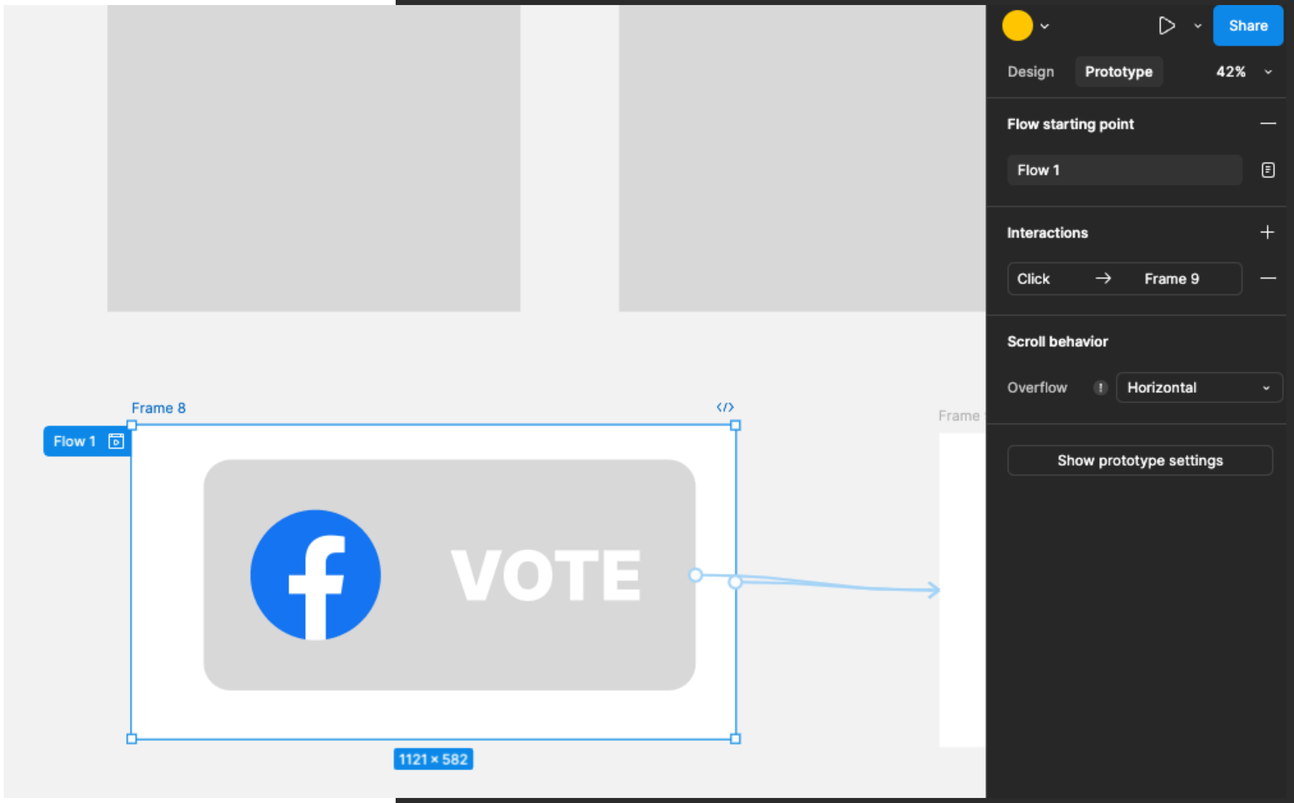
Prototyping

1. Create Flow
2. Set Device
3. Add overflow behaviour „scrolling“
4. Click on „Play“ button



Prototyping

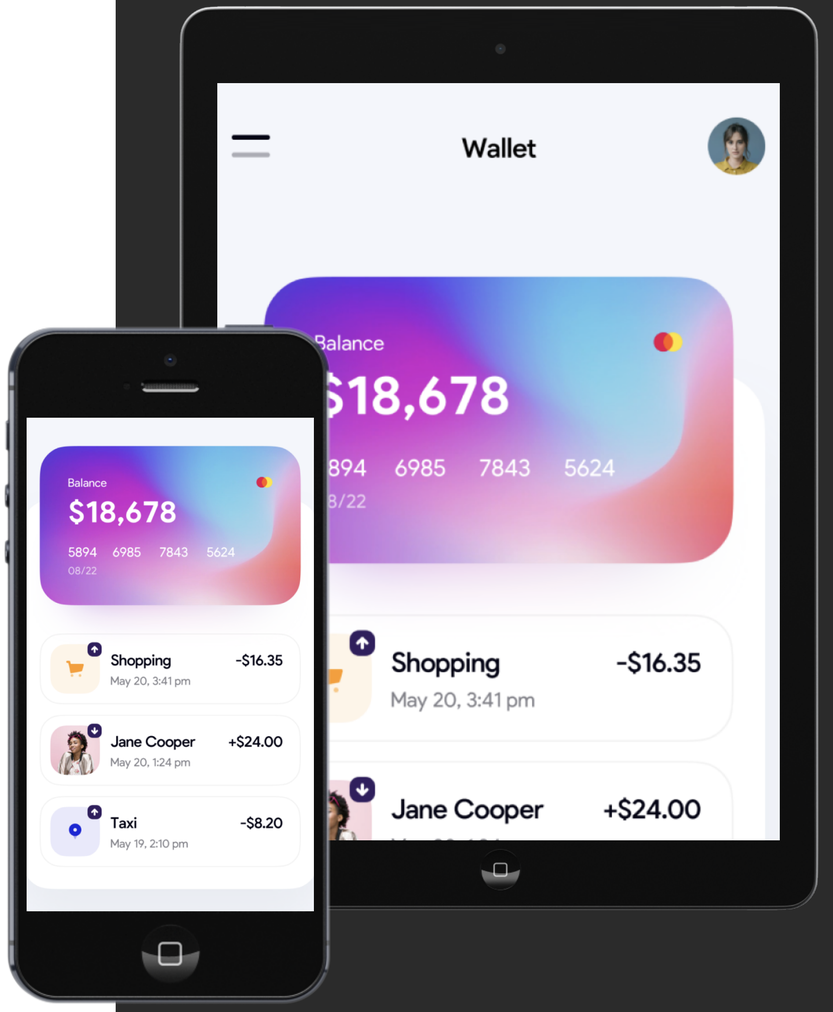
- In properties menu select **interaction** (swipe, click, mouseover etc.) and type of **animation** (move in, push in, slide out etc.)
- Overflow behavior: Besides vertical scrolling Figma offers realistic overflow behavior



User Tests

Show prototypes on your phone with the app „Figma mirror“

- Use the mirror view of the mobile app
- View real-time edits to your design
- Preview design scaled to device screen

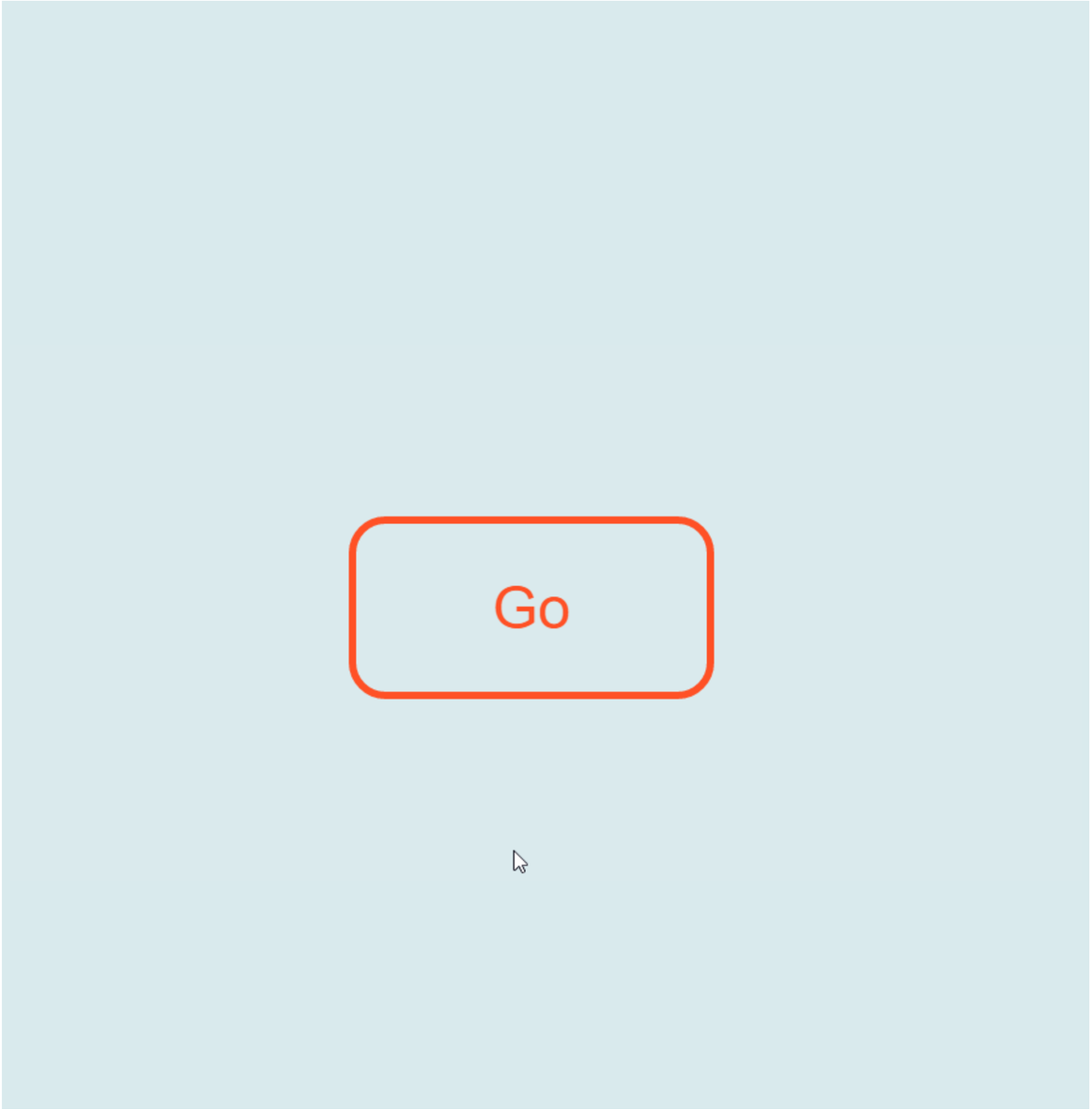


Animations

Animation in UX must be unobtrusive, brief, and subtile.

Use it for:

- ✓ feedback
- ✓ state-change
- ✓ navigation metaphors
- ✓ to enhance signifiers



Animations

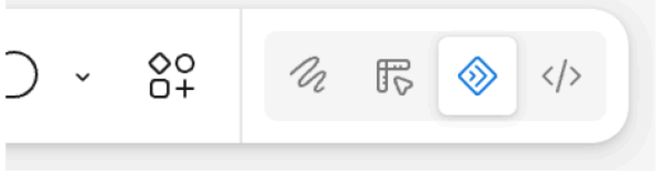
.. should be fast, precise and only on area not whole page

- ✓ Use **intentionally**, not decoratively
- ✓ Control: enable **pause, stop or hide** for auto-playing animations (e.g. banner, slider) and moving/blinking content; when there are many animation possibility to pause **whole page**
- ✓ Reduce: when the OS describes someone wants less motion, the interface has to react



Figma Motion

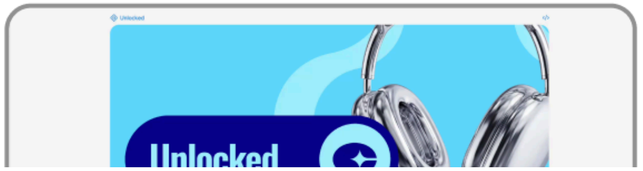
1. Switch to the Motion mode.



2. Select the layer with the Motion icon.



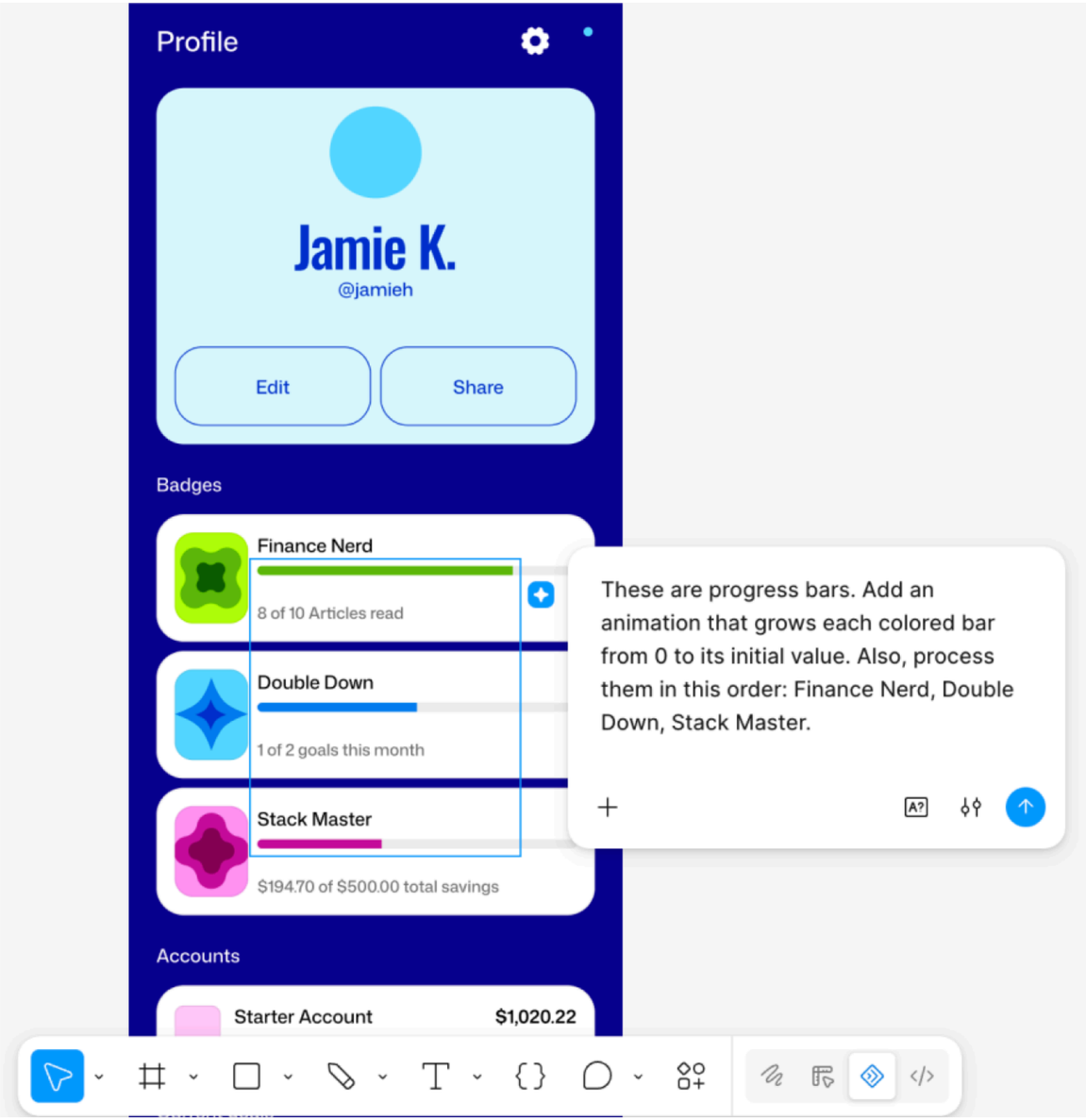
3. Hit the spacebar and let it roll.



Figma Motion

Describe what you want, and the **agent** builds real keyframes on the timeline.

Fine-tune by hand or keep prompting.



TOOL	GENERATE UI	LOW-FI SUITABLE	WIREFLOW	BEST / WORST USED
Stitch	★★★	★★		+ exploring screens and user journeys
Lovable	★★★	★		+ Functional apps with real logic & data – less suited to rough exploration
Figma Make	★★★	★	★	+ functional, Interactive prototypes – Quickly becomes polished / code-based

attach visual reference/library

bullet points for readability

add what (not) to do

first prompt is important, then iterate

Erstelle einen responsiven, interaktiven Registrierungs-Flow.

Nutze mein hochgeladenes Design System für Farben, Typografie, Abstände und Komponenten.

Layout

- Links: Login Bereich
- Rechts: übersichtliches Formular mit großzügigem Weißraum und Abbildung.

Registrierung Szenario

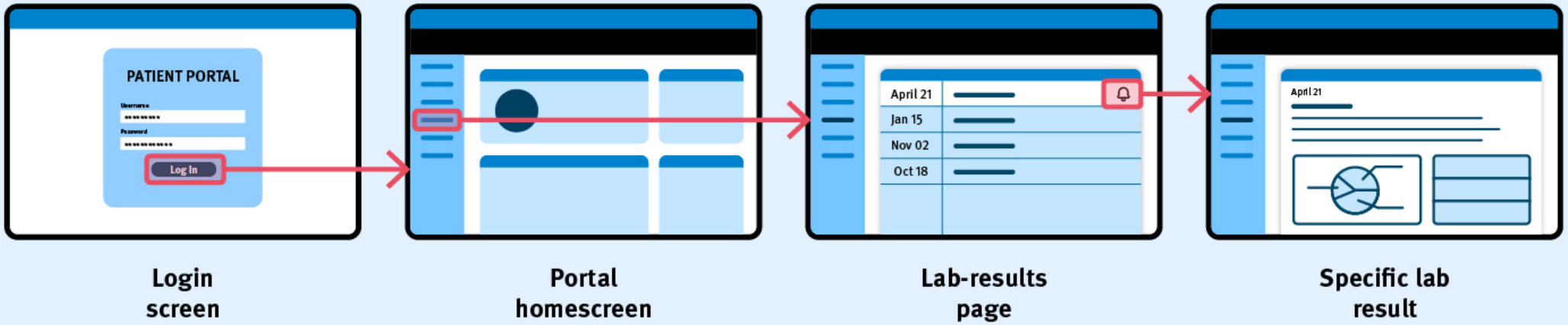
1. E-Mail eingeben oder mit Google/Apple fortfahren.
2. Passwort erstellen, inklusive Ein-/Ausblenden und verständlichen Anforderungen.
3. Optionalen Anzeigenamen ergänzen, danach Erfolgsansicht.

Zeige den aktuellen Schritt und biete Zurücknavigation ohne Datenverlust. Google/Apple überspringen den Passwortschritt. Ergänze „Schon ein Konto? Anmelden“.

Anweisungen

- Deutsche UI-Texte mit „Du“-Ansprache.
- Klare Fehlermeldungen sowie Lade- und Erfolgszustände.
- Tastaturbedienbarkeit, sichtbare Fokuszustände und ausreichende Kontraste.
- Mobil: einspaltig, Formular im Fokus.
- Vollständig durchklickbar; Registrierung und Social Login nur simulieren.

User Flow



Wireflows document the ideal or typical **pathway** through screens that need to be visited in order to accomplish a given **task**

👋 Hands on

1) High-Fi Wireflow

Wireflow: create flow with AI showing **one core functionality,** use your **storyboard** as base

Happy Path = Show how the user goes through system in ideal flow. everything works as expected - the user finishes his goal

Keep prototype simple and focused on design and interaction, attach a **Design Library**

Visualise the interaction 

2) Critique

- ✓ What did AI assume?
- ✓ What was not defined in the requirements?
- ✓ Which UX decisions did the AI make for us?
- ✓ Does the flow actually solve the user problem?
- ✓ What would we change?

Best Practices: Creating Prototypes

- ✓ **Beware of the principles of visual design:** Balance: use grids, Contrast: Use a Contrast Checker.
- ✓ **Plan your user flows:** Before you start prototyping sketch out the flow on paper.
- ✓ **Use Components and Styles:** Figma's component system allows you to create reusable UI elements. This ensures consistency and makes it easier to update design elements throughout the prototype. A Design System helps to maintain consistency.
- ✓ **Version Control:** Figma has version control built-in, which allows you to track changes over time. Use this feature to keep track of design iterations and easily roll back to a previous version if needed.

Best Practices: Creating Prototypes

- ✓ **User-Centered Design:** Always keep the end-user in mind. Your prototype should reflect the needs and expectations of your target audience.
- ✓ **User Testing:** Share your prototype with stakeholders or potential users for feedback. Figma allows to create shareable links, so others can interact with your prototype.
- ✓ **Iterate and Refine:** Use the feedback you gather to iterate and refine your prototype. The goal is to create a design that meets user needs.
- ✓ **Collaboration:** Figma is a collaborative tool, so make use of its collaboration features with your team members.

Best Practices: Creating Prototypes

- ✓ **Microinteractions, Transitions and Animations:** Don't forget about microinteractions, animations or transitions, that add depth to your prototype, e.g. hover effects, loading spinners, or page transitions. You can set the duration, easing, and delay for each transition to make them feel natural. Add interactive elements e.g. buttons and forms.
- ✓ **Responsive Design:** Design your prototype with responsiveness in mind. Figma allows you to create designs for various screen sizes and easily switch between them.
- ✓ **Organize:** Keep your Figma file organized by using frames to represent different screens or sections. Name layers and frames so that others can understand.

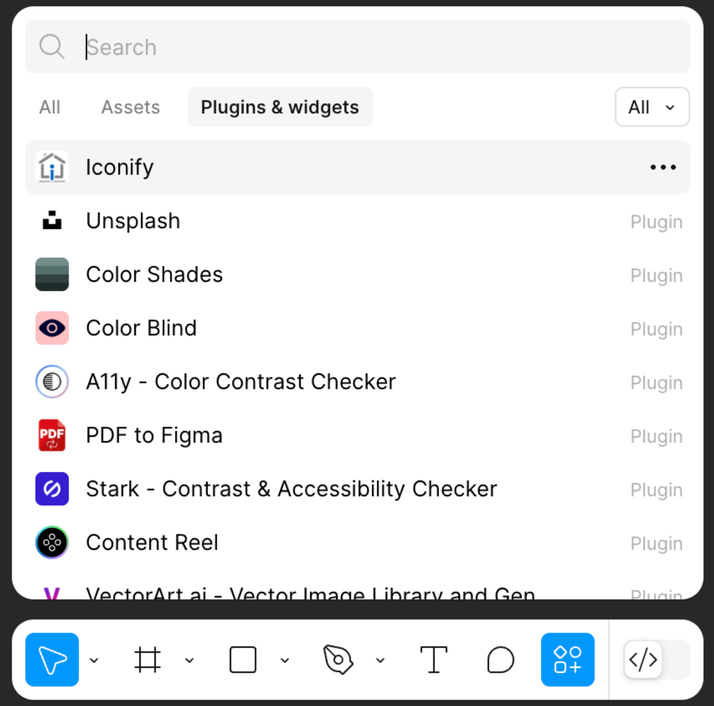
Figma Community

Figma Community: https://www.figma.com/community

Plugins

- Color Contrast Checker, AI Generators, Icon Library,..
- Unsplash: Images
- Lottie for Animations

Design Systems like Uber, Teams, React, ..



UX management & design workflow

UX management



... managing UX across product process to systematically integrate user needs into UX strategy, design decisions, agile development and evaluation

Translates **research insights** into user stories, requirements & design decisions

Defines **UX goals, metrics & targets** to evaluate whether the experience works

Connects UX work with **agile** development to consider user needs

Responsibilities

VS

UX

User requirements

Research

Design

UX roadmap

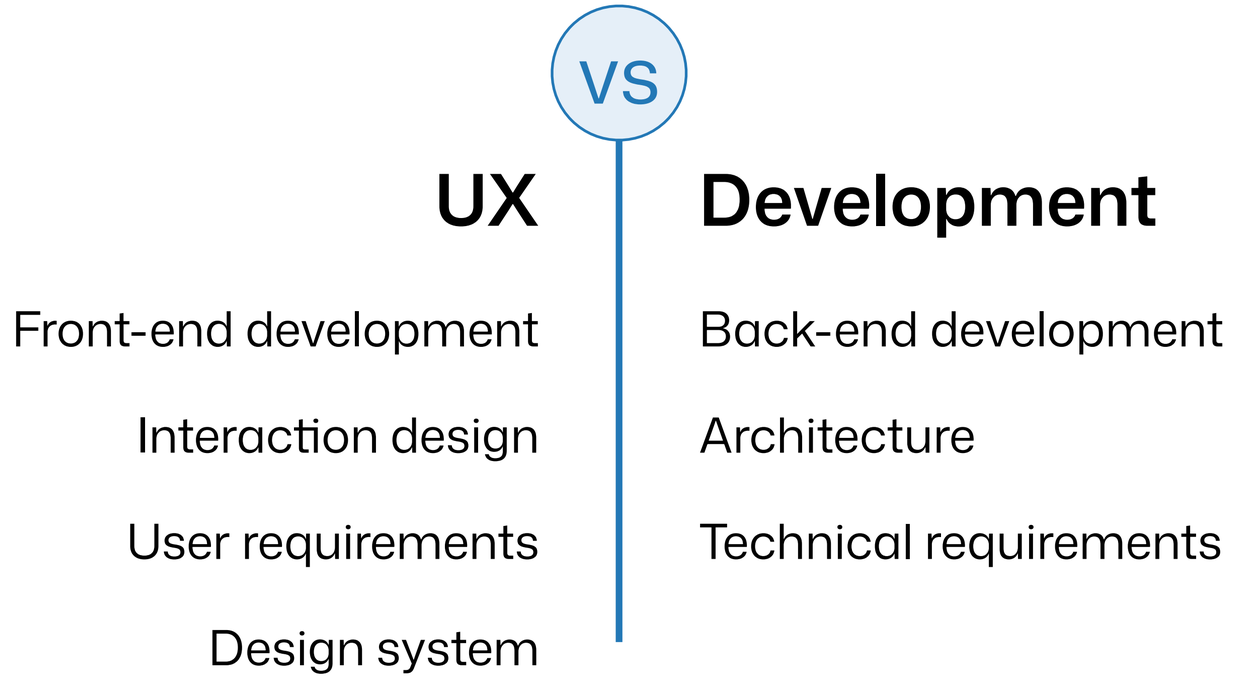
Product

Business requirements

Market requirements

Product roadmap

Responsibilities



Responsibilities

VS

UX

User requirements

Design system

Usability studies

Marketing

Branding guidlines

Market landscape

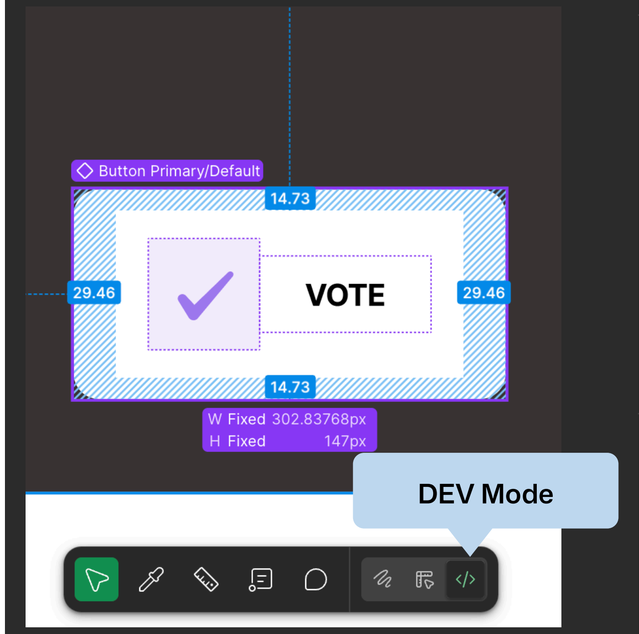
Market surveys

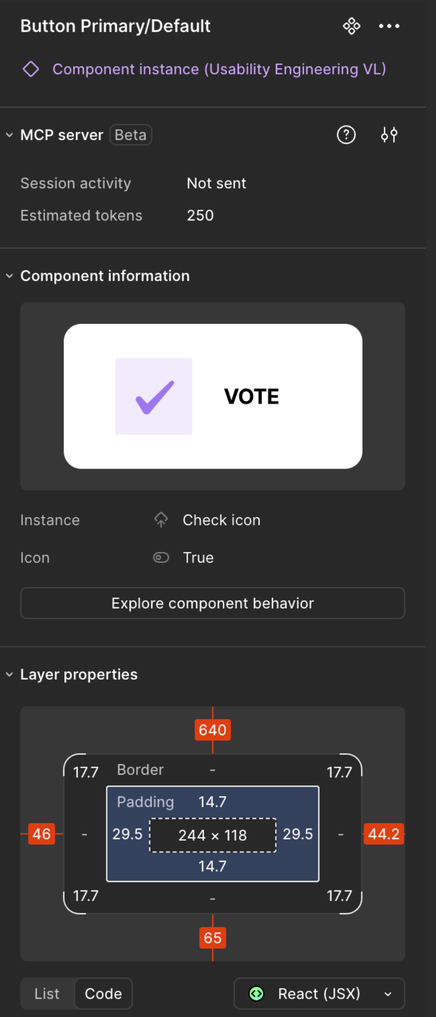
Development <> Design

Challenge: Translating between design and code

👉 Goal: Fewer misunderstandings, faster implementation

- ✓ Shared components & good design documentation
- ✓ Teach developers measurements & component states
- ✓ Tools for linking (e.g. design ↔ code) or/and storybook
- ✓ Defined tokens and fonts



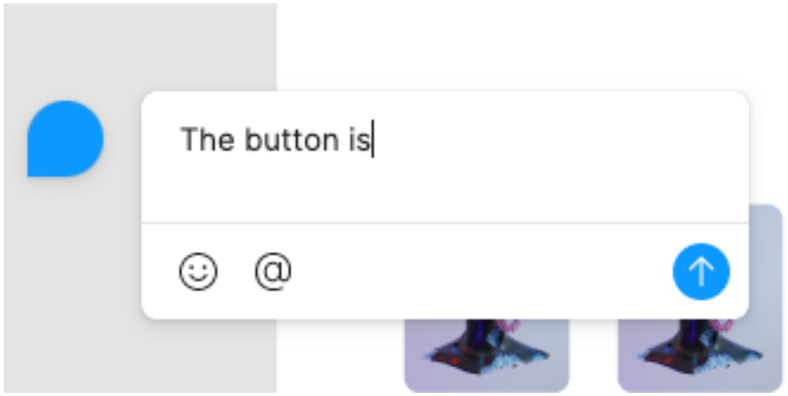


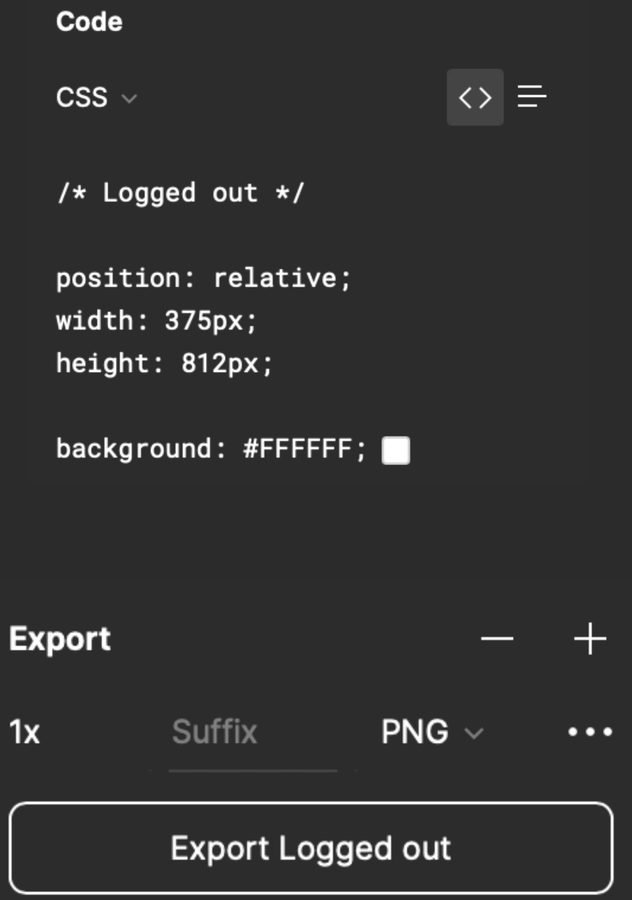
Development <> Design

✓ **Export** layers as PNG, JPG, SVG, PDF

✓ **Code exports** (CSS, Angular, React) for frontend development possible with figma

✓ **Comments** for team collaboration





Documentation & Handover

- Updated regularly
- Explain design decisions and use cases
- Define ownership and update process
- Act as single source of truth
- Support consistency and reusing
- Include components, code snippets & design rules

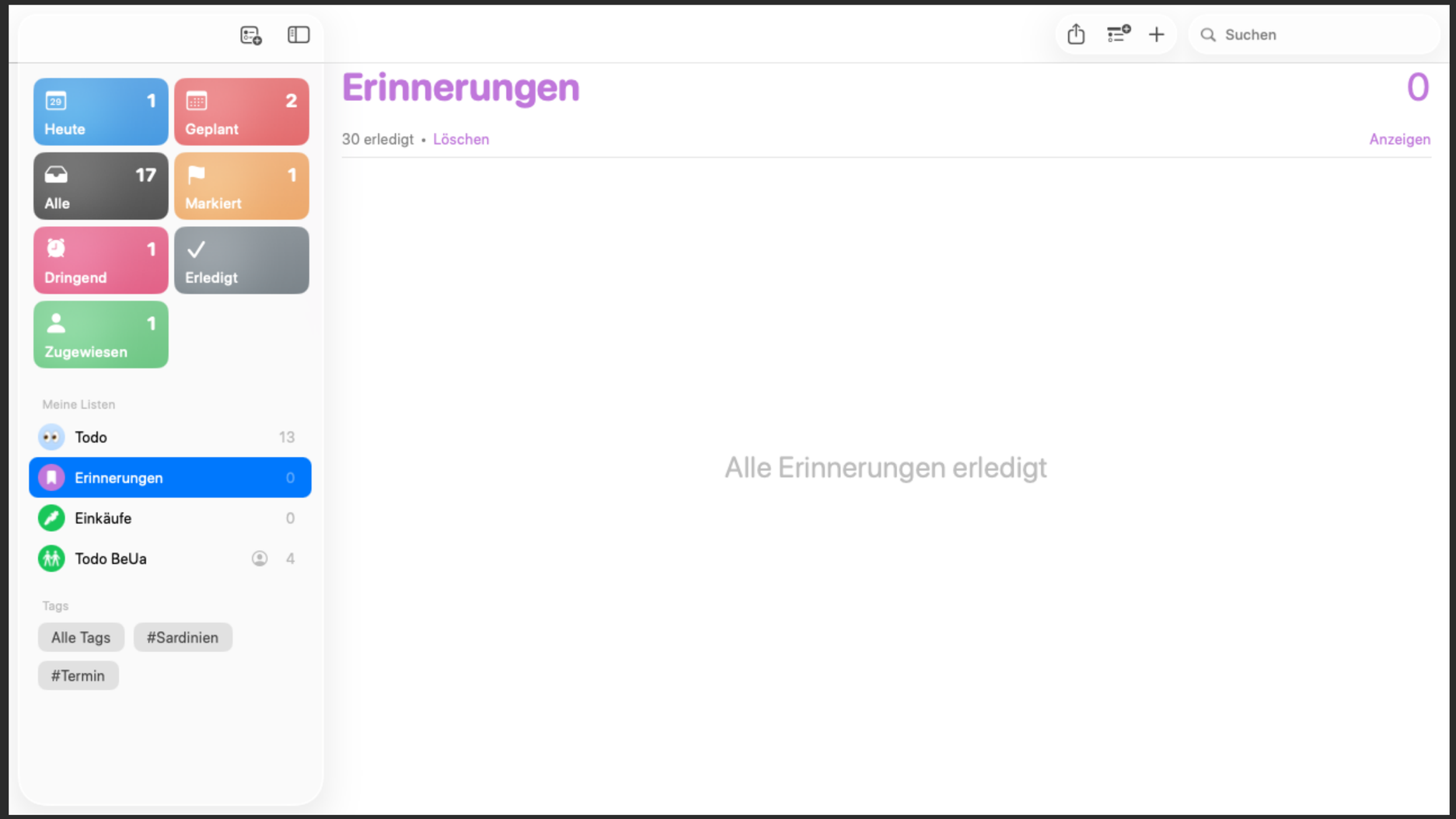
Feedback

- ✓ Specific
- ✓ Include outcome

- X Don't accuse
- X Don't take someone else's interpretation



🤔 Design Review

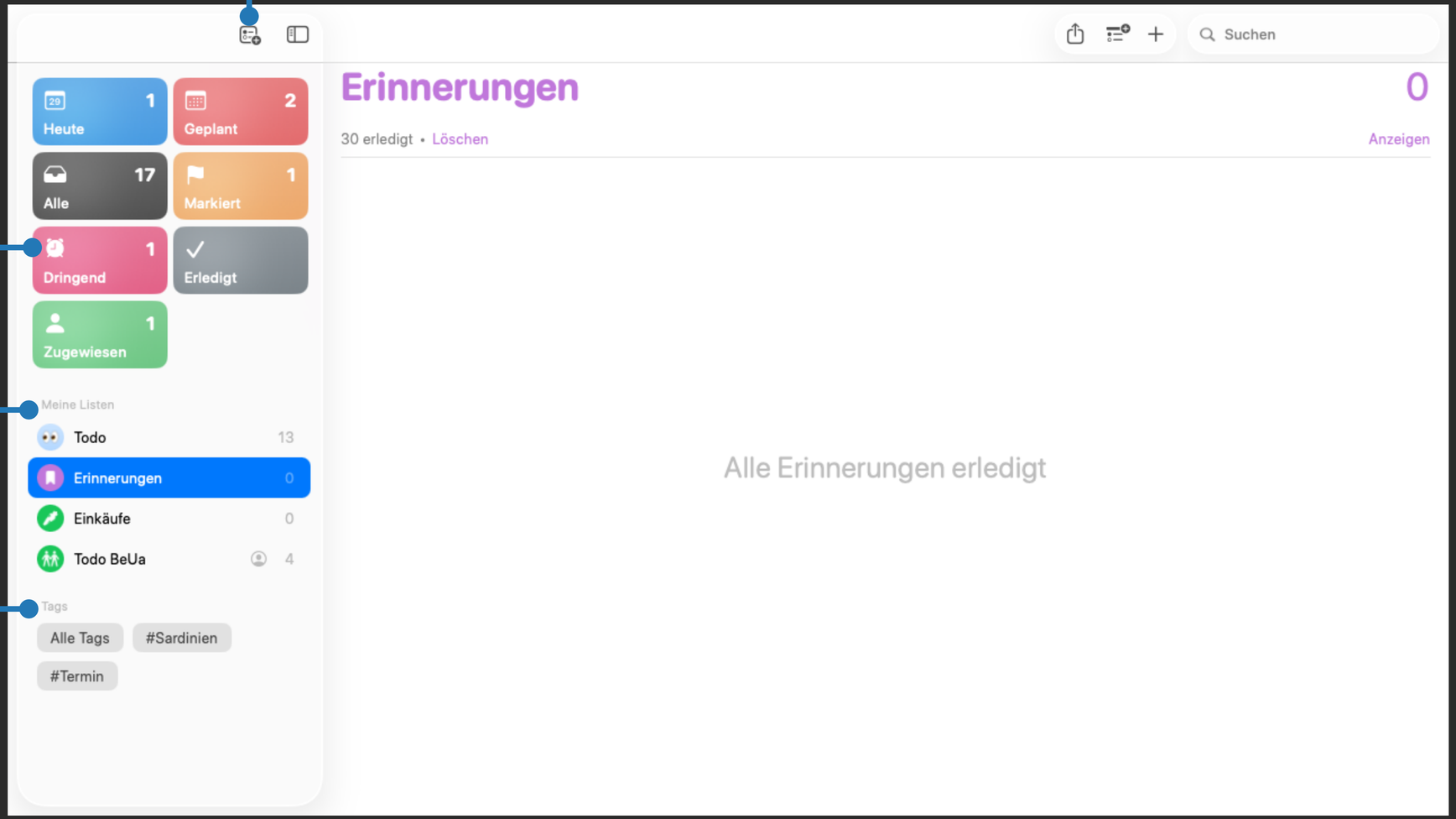


Button not next to list (proximity)

Not important but very prominent

Too light (Accessibility?)

Why do I need tags?



References

Hartson, R & Pyla, P 2019, The UX book: Agile UX design for a quality user experience, Elsevier Inc, Cambridge, MA, Science Direct

Lidwell, W., Holden, K., & Butler, J. (2010). *Universal principles of design*. Rockport Publishers.

Staiano, F.,Designing and Prototyping Interfaces with Figma Learn essential UX_UI design principles by creating interactive prototypes for mobile, tablet, and desktop, 2022

Twarog, A., & Moller, G. (2024). *Design for developers*. Independently published.